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Literary/ Dramatic on

“A novel speech enhancement method for noise reduction Using Modified LMS Algorithm”

Aim:

To develop Modified Least Mean Square (MLMS) algorithm for speech de-noising method is tested on various type of noise.

Objectives:

The comprehensive scope prior to the technique has been explained by the literature review, especially for speech enhancement techniques. The primary challenge ahead in developing a technique for Modified Least Mean Square (MLMS) is to accomplish remarkable speech Enhancement. Hence, we define the main objectives of our research work as follows:

- To construct a speech database of different speech signal.
- To formulate and develop a Modified Least Mean Square (MLMS) technique for a speech Enhancement.
- To proposed de-noising algorithm is tested with standard speech signal with various types of noises (i.e., Airport, Babble, Restaurant and AWGN noises) are used at different levels (i.e., 0 dB, 5 dB, 10 dB and 15 dB) to analyse the performance of the proposed algorithm

Problem Statement of Proposed System:

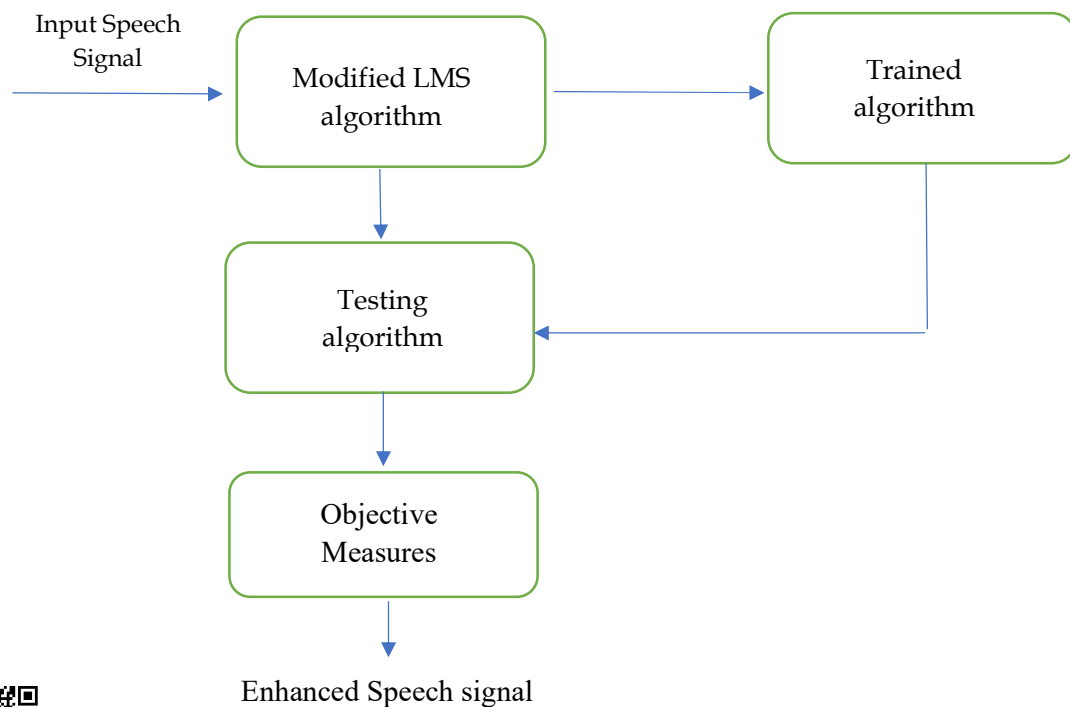


Fig 1: Block Diagram of Proposed System



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Step-by-Step Algorithm

Step1: Noisy speech signal is applied to the modified Least Mean Square algorithm (MLSM).

Step2: In MLSM, initially set the weights=0 and choose the step size, orthogonal matrix.

Step3: For each time step compute transfer domain input.

Step4: Compute the filter output and error.

Step5: In order to determine the optimal filter coefficients, the training coefficient, which updates the filter coefficient values in the MLMS algorithm, is multiplied by the current input values, which represent the instantaneous gradient.

Step6: We must set the gradient value to zero in order to minimize the mean square error as much as possible.

Step7: Compute objective measures namely, dBSNR, LLR, and ISD.

Advantages.

- Especially under uncorrelated inputs modified LMS algorithm works faster convergence.
- Improve better control of step size and normalization.
- Modified LMS algorithm works efficiently for long adaptive filters.

Applications.

- Noise removal in ECG or EEG signals.
- Speech enhancement in mobile communication.
- Active noise cancellation in headphones.
- Adaptive Equalization in Hearing aid devices.

Conclusion

The performance of the adaptive filter using the MLMS algorithm is evaluated, and a modified LMS algorithm is proposed. The suggested approach, which adaptively filters input signals in both the frequency and time domains, is improved using the diffusion least mean-square algorithm to produce a good de-noise algorithm. According to the three objective measures—dBSNR, LLR, and ISD—the current experiment demonstrated that the suggested approach did a good job of enhancing the speech signal's quality over the LMS Algorithm. Future research should concentrate on examining the suggested method's performance in various operating situations, including noisy environments, by combining the goal of assessing the suggested algorithm's performance.



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